Manish Rastogi

Senior Software Engineer — Backend & AI Systems

Bengaluru, India · +91 960 193 6216 · manish.rastogi94@gmail.com · manishrastogi.me · github.com/manish001in · linkedin.com/in/manish--rastogi

SUMMARY

Backend and AI engineer, 10+ years, building systems end to end — from low-latency trading platforms to LLM agents in production. Most recently built the provider-agnostic tool-calling architecture and the full integration layer for an AI assistant, plus the evals that measured whether the agents actually worked. Comfortable owning a problem from architecture through on-call, have led teams up to ~10, and spent two years founding my own product — so I bias toward shipping and cost, not just design docs.

SKILLS

Languages: Python, JavaScript / TypeScript, Go
Backend: Django, FastAPI, Flask, Node.js
AI / LLM: Tool-calling agents, evals, embeddings & retrieval, RAG · OpenAI, Anthropic, Gemini · LlamaIndex, LangGraph
Data & infra: PostgreSQL, MySQL, Redis, Supabase · AWS (EC2, ECS, S3, Lambda, RDS), Modal, Docker, GitHub CI

EXPERIENCE

Studio Management (Littlebird) Aug 2025 – Jul 2026

Senior Software Engineer · Bengaluru

Littlebird, an AI executive assistant (grew from 250 to 5,000 users). Also built Songbird, a music-recommendation product.

- Rebuilt the agent's core tool-calling loop into a **provider-agnostic plugin system** (OpenAI Responses + Gemini) with per-turn tool management — the pattern other engineers then built their own agents on.
- Owned the **integrations layer end to end, backend through UI**: Composio and OAuth with per-tool routing on the backend, plus the toggle wired across the client SDK, stores, and UI — connecting **8 platforms** (Gmail, Drive, Docs, Sheets, Calendar, Linear, GitHub, Notion).
- Built the **evaluation framework** (judge prompts and datasets) for the main agent and subagents, raising integration eval scores **from 64% to 85%+** and catching regressions before release.
- Built the **meeting-notes backend**: turned in-app meeting transcripts into cleanly formatted notes with speaker handling, and matched Google Calendar events to the meetings a user started so each note attached to the right meeting.
- Drove **Google's CASA security certification** for the app end to end — the review required for production access to users' Gmail, Drive, and Calendar data.
- Added the **reliability and observability** behind it: usage and error telemetry with dashboards, plus handling for dropped connections and interrupted model streams.

Songbird — solo build *(music recommendation)*

- Designed and built the **entire platform alone** — data pipeline, infrastructure on Modal, and the recommendation modeling.
- **Modeled each user's taste** from their listening history: enriched every track with LLM calls (Gemini with search grounding) and audio features, turned them into **embeddings**, then reduced dimensions two ways as Modal jobs — unsupervised (PCA) to visualize a library, and supervised (PLS) aligned to the user's own likes and dislikes.
- Built a **preference model** that scores each song's appeal from **~100 pieces of feedback** (hearts, thumbs, ranked comparisons), calibrated so its head-to-head predictions stay honest — refreshing as new ratings arrive and driving what gets recommended next.
- Re-architected **from Spotify-only to provider-agnostic**, adding **Apple Music** behind the same interface.

Docsum Feb 2025 – Jun 2025

Software Engineer (Contract) · Bengaluru

- Built the **async task system** (an async-tasks table with a streaming status endpoint) powering **bulk document operations** — bulk archive and bulk reprocess — across high-volume workloads.
- Built the **document-relationship layer**: new relationship tables plus a bulk endpoint that computes relationships across many documents at once, forming the backbone of the **inter-document knowledge graph** (pgRouting + LLMs).
- Built the **Google Drive integration** (Paragon workflows) and file-sync APIs for bringing external documents in.
- **Hardened the FastAPI / SQLAlchemy backend**: fixed dependency-injection and DB-session bugs, file-type detection, and persistence issues.

Tech: Python, FastAPI, SQLAlchemy, Hatchet, PostgreSQL

Twing Dec 2022 – Jan 2025

Founder · India

Solo-founded an AI productivity startup; built and shipped four products.

- **TwingBot —** an AI Slack bot for task management; **~200 users, saving 30+ minutes a day**.
- **Legaling —** RAG-based legal research and case management (Azure Vector DB, OpenAI, Search Service).
- **TwingParse —** template-based extraction from varied documents using OCR and language detection.
- **TwingTranslate —** translation for Indian languages across scanned and digital documents.

Tech: Python, FastAPI, PostgreSQL, Celery, AWS, Azure, LlamaIndex, LangChain

Salarybox Apr 2022 – Jul 2022

Technical Lead · Gurugram

- Built a **real-time notification engine** from scratch that scaled to **3,000 notifications/min** for thousands of simultaneous users, using Redis, queues, and batching over a dynamic rule set.
- Cut **latency from 30s to 1–2s** and **DB locks from ~300 to 2–5 per day** by fixing several problem processes; added monitoring and the ability to re-run requests in testing.
- Cut **AWS and infrastructure spend ~30%** by moving to ARM servers and rightsizing after performance tests.
- Led and mentored a **team of 6**.

Tech: Python, Django, Redis, PostgreSQL, Celery, AWS

Instawork Oct 2021 – Apr 2022

Senior Software Engineer

- Built the **tipping feature** (multiple touchpoints, rating-based calculation) that raised client tipping **by up to 15%**.
- Added **fraud-detection checks** that analyzed tipping behavior against expected parameters.

Tech: Python, Django, JavaScript

Quadeye Securities Feb 2017 – Dec 2020

Software Developer · Gurgaon

- Architected a **low-latency trading-visualization app** handling **up to 3,000 data points/second**, with live alerts and commands into trading strategies.
- Drove **~5x throughput gains** — websockets, Redis caching, and queue-based offloading — taking capacity **from 5k to 1M events/day**.
- Helped hire and manage a **team of 10+**.

Tech: Python, Django, jQuery, JavaScript, MySQL, Redis

Ishi Systems Dec 2015 – Jan 2017

Software Developer · Ahmedabad

- Built **on-the-fly video processing** (Intel Media SDK, Go), made systems **ONVIF-compliant**, and worked full-stack across Python / Django and AngularJS.

PROJECTS

Clausewize *(clausewize.app · solo)*

- A Chrome extension that turns terms-of-service and privacy pages (and PDFs) into a plain **risk brief**: a verdict, severity-rated clauses, and the exact clause each flag came from. **Bring-your-own-key across OpenAI, Anthropic, and Gemini**; privacy-first (stores no raw documents); with an account tier for synced history and shareable briefs. Live on the Chrome Web Store.

Pushbridge *(open source, 18★ · github.com/manish001in/pushbridge)*

- A **Manifest V3 Chrome extension** that rebuilds core Pushbullet features (notification mirroring, SMS/MMS, push and file sync) on the official REST and WebSocket APIs. TypeScript + Lit, unit and end-to-end tested, 12 releases. The hard part: keeping a live WebSocket sync working under MV3's short-lived service worker.

EDUCATION

Nirma University — B.Tech, Information Technology 2012 – 2016